How-To Library

Informatica

Secure@Source Database Requirements

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Abstract

This article summarizes the privileges that are required to install Secure@Source and to run scan jobs.

Supported Versions

- Secure@Source 3.x

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Overview

This article summarizes the databases and database user privileges that are required for the Secure@Source installation and the user privileges that are required to run a scan job.

Secure@Source works with other services in the Informatica domain. Before you create the application services, you must verify the database requirements and set up the databases and database user accounts.

Secure@Source requires additional user account privileges when you run a scan job. You must configure these privileges before you run a scan on a data store.

For information about Secure@Source installation, see the *Secure@Source Installation and Configuration Guide*. For information about running scan jobs, see the *Secure@Source Administrator and User Guide*.

Prepare the Databases for the Informatica Domain Overview

Informatica stores data and metadata in repositories in the domain. Before you create the domain and the application services, set up the databases and database user accounts for the repositories.

Set up a database and user account for the following repositories:

- Domain configuration repository

- Model repository
- Profiling warehouse
- Reference data warehouse
- Secure@Source repository

To prepare the databases, verify the database requirements and set up the database. The database requirements depend on the application services that you create in the domain and the number of data integration objects that you build and store in the repositories.

Set Up Database User Accounts

Set up a database and user account for the domain configuration repository and for the repository databases associated with the applications services.

Use the following rules and guidelines when you set up the user accounts:

- The database user account must have permissions to create and drop tables, indexes, and views, and to select, insert, update, and delete data from tables. The users must have sufficient tablespace sizes.
- Use 7-bit ASCII to create the password for the account.
- To prevent database errors in one repository from affecting any other repository, create each repository in a separate database schema with a different database user account. Do not create a repository in the same database schema as the domain configuration repository or any other repository in the domain.
- If you create more than one domain, each domain configuration repository must have a separate user account.

You must create the following database user accounts:

- Secure@Source repository
- Model repository
- Domain configuration repository
- Content Management
- Profiling warehouse

Domain Configuration Repository Database Requirements

Informatica components store metadata in relational database repositories. The domain stores configuration and user information in a domain configuration repository.

You must set up a database and user account for the domain configuration repository before you run the installation. The database must be accessible to all gateway nodes in the Informatica domain.

When you install Secure@Source, you provide the database and user account information for the domain configuration repository. The Secure@Source installer uses JDBC to communicate with the domain configuration repository.

The domain configuration repository supports the following database types:

- IBM DB2 UDB
- Microsoft SQL Server
- Oracle
- Sybase ASE

Allow 200 MB of disk space for the disk space.

IBM DB2 Database Requirements

Use the following guidelines when you set up the repository on IBM DB2:

- If the repository is in an IBM DB2 9.7 database, verify that IBM DB2 Version 9.7 Fix Pack 7 or a later fix pack is installed.
- On the IBM DB2 instance where you create the database, set the following parameters to ON:
  - DB2_SKIPINSERTED
  - DB2_EVALUNCOMMITTED
  - DB2_SKIPDELETED
  - AUTO_RUNSTATS
- On the database, set the configuration parameters.

  The following table lists the configuration parameters that you must set:

Parameter	Value
applheapsz	8192
appl_ctl_heap_sz	8192
logfilsiz	8000
maxlocks	98
locklist	50000
auto_stmt_stats	ON

- Set the tablespace pageSize parameter to 32768 bytes.

  In a single-partition database, specify a tablespace that meets the pageSize requirements. If you do not specify a tablespace, the default tablespace must meet the pageSize requirements.

  In a multi-partition database, specify a tablespace that meets the pageSize requirements. Define the tablespace in the catalog partition of the database.
- Set the NPAGES parameter to at least 5000. The NPAGES parameter determines the number of pages in the tablespace.
- Verify that the database user has CREATETAB, CONNECT, and BINDADD privileges.
- Informatica does not support IBM DB2 table aliases for repository tables. Verify that table aliases have not been created for any tables in the database.
- In the DataDirect Connect for JDBC utility, update the DynamicSections parameter to 3000.

  The default value for DynamicSections is too low for the Informatica repositories. Informatica requires a larger DB2 package than the default. When you set up the DB2 database for the domain configuration repository or a Model repository, you must set the DynamicSections parameter to at least 3000. If the DynamicSections parameter is set to a lower number, you can encounter problems when you install or run Informatica services.

Microsoft SQL Server Database Requirements

Use the following guidelines when you set up the repository on Microsoft SQL Server:

- Set the read committed isolation level to READ_COMMITTED_SNAPSHOT to minimize locking contention.
  To set the isolation level for the database, run the following command:

  ```
  ALTER DATABASE DatabaseName SET READ_COMMITTED_SNAPSHOT ON
  ```

  To verify that the isolation level for the database is correct, run the following command:

  ```
  SELECT is_read_committed_snapshot_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the ALLOW_SNAPSHOT_ISOLATION parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET ALLOW_SNAPSHOT_ISOLATION ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_allow_snapshot_isolation_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the AUTO_UPDATE_STATISTICS_ASYNC parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS_ASYNC ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_async_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the AUTO_UPDATE_STATISTICS parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_on FROM sys.databases WHERE name = DatabaseName
  ```

- The database user account must have the CONNECT, CREATE TABLE, and CREATE VIEW privileges.

Oracle Database Requirements

Use the following guidelines when you set up the repository on Oracle:

- Set the open_cursors parameter to 4000 or higher.
- Set the permissions on the view `$parameter` for the database user.
- Set the privileges for the database user to run *show parameter open_cursors* in the Oracle database.
  When you run the pre-installation (i10Pi) system check tool, i10Pi runs the command against the database to identify the OPEN_CURSORS parameter with the domain database user credentials.

  You can run the following query to determine the open cursors setting for the domain database user account:

  ```
  SELECT VALUE OPEN_CURSORS FROM V$PARAMETER WHERE UPPER(NAME)=UPPER('OPEN_CURSORS')
  ```

- Verify that the database user has the following privileges:

  CREATE SEQUENCE

  CREATE SESSION

  CREATE SYNONYM

  CREATE TABLE

  CREATE VIEW

- Informatica does not support Oracle public synonyms for repository tables. Verify that public synonyms have not been created for any tables in the database.

Sybase ASE Database Requirements

Use the following guidelines when you set up the repository on Sybase ASE:

- Set the database server page size to 16K or higher. You must set the page size to 16K as this is a one-time configuration and cannot be changed afterwards.
- Set the database locking configuration to use row-level locking.
  The following table describes the database locking configuration that you must set:

Database Configuration	Sybase System Procedure	Value
Lock scheme	sp_configure "lock scheme"	0, datarows

- Set the Sybase database option "ddl in tran" to TRUE.
- Set "allow nulls by default" to TRUE.
- Turn ON the Sybase database option select into/bulkcopy/pllsort.
- Enable the "select" privilege for the sysobjects system table.
- Create the following login script to disable the default VARCHAR truncation:

```
create procedure dbo.sp_string_rtrunc_proc as set string_rtruncation on
sp_modifylogin "user_name", "login script", sp_string_rtrunc_proc
```

  The login script is executed every time the user logs into the Sybase instance. The stored procedure sets the parameter at the session level. The sp_modifylogin system procedure updates "user_name" with the stored procedure as its "login script". The user must have permission to invoke the stored procedure.

- Verify that the database user has CREATE DEFAULT, CREATE PROCEDURE, CREATE RULE, CREATE TABLE, and CREATE VIEW privileges.
- Set the database configurations to the recommended baseline values.
  The following table lists the database memory configuration parameters that you must set:

Database Configuration	Sybase System Procedure	Value
Maximum amount of total physical memory	sp_configure "max memory"	2097151
Procedure cache size	sp_configure "procedure cache size"	500000
Number of open objects	sp_configure "number of open objects"	5000
Number of open indexes	sp_configure "number of open indexes"	5000
Number of open partitions	sp_configure "number of open partitions"	5000
Heap memory per user	sp_configure "heap memory per user"	49152
Number of locks	sp_configure "number of locks"	100000

Data Object Cache Database Requirements

The data object cache database stores cached logical data objects and virtual tables for the Data Integration Service. You specify the data object cache database connection when you create the Data Integration Service.

The data object cache database supports the following database types:

- IBM DB2 UDB
- Microsoft SQL Server
- Oracle

Allow 200 MB of disk space for the database.

Note: Ensure that you install the database client on the machine on which you want to run the Data Integration Service.

IBM DB2 Database Requirements

Use the following guidelines when you set up the repository on IBM DB2:

- Verify that the database user account has CREATETAB and CONNECT privileges.
- Informatica does not support IBM DB2 table aliases for repository tables. Verify that table aliases have not been created for any tables in the database.
- Set the tablespace pageSize parameter to 32768 bytes.
- Set the NPAGES parameter to at least 5000. The NPAGES parameter determines the number of pages in the tablespace.

Microsoft SQL Server Database Requirements

Use the following guidelines when you set up the repository on Microsoft SQL Server:

- Set the read committed isolation level to READ_COMMITTED_SNAPSHOT to minimize locking contention. To set the isolation level for the database, run the following command:

  ```
  ALTER DATABASE DatabaseName SET READ_COMMITTED_SNAPSHOT ON
  ```

  To verify that the isolation level for the database is correct, run the following command:

  ```
  SELECT is_read_committed_snapshot_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the ALLOW_SNAPSHOT_ISOLATION parameter to ON. To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET ALLOW_SNAPSHOT_ISOLATION ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_allow_snapshot_isolation_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the AUTO_UPDATE_STATISTICS_ASYNC parameter to ON. To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS_ASYNC ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_async_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the AUTO_UPDATE_STATISTICS parameter to ON. To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_on FROM sys.databases WHERE name = DatabaseName
  ```

- The database user account must have the CONNECT and CREATE TABLE privileges.

Oracle Database Requirements

Use the following guidelines when you set up the repository on Oracle:

- Verify that the database user has the following privileges:

  CREATE INDEX

  CREATE SESSION

  CREATE SYNONYM

  CREATE TABLE

  CREATE VIEW

  DROP TABLE

  INSERT INTO TABLE

  UPDATE TABLE

- Informatica does not support Oracle public synonyms for repository tables. Verify that public synonyms have not been created for any tables in the database.

Model Repository Database Requirements

Secure@Source services and Informatica clients store data and metadata in the Model repository. Before you create the Model Repository Service, set up a database and database user account for the Model repository.

The Model repository supports the following database types:

- IBM DB2 UDB
- Microsoft SQL Server
- Oracle

Allow 3 GB of disk space for DB2. Allow 200 MB of disk space for all other database types.

For more information about configuring the database, see the documentation for your database system.

IBM DB2 Database Requirements

Use the following guidelines when you set up the repository on IBM DB2:

- If the repository is in an IBM DB2 9.7 database, verify that IBM DB2 Version 9.7 Fix Pack 7 or a later fix pack is installed.
- On the IBM DB2 instance where you create the database, set the following parameters to ON:
  - DB2_SKIPINSERTED
  - DB2_EVALUNCOMMITTED
  - DB2_SKIPDELETED
  - AUTO_RUNSTATS
- On the database, set the configuration parameters.

The following table lists the configuration parameters that you must set:

Parameter	Value
applheapsz	8192
appl_ctl_heap_sz	8192
logfilsiz	8000
maxlocks	98
locklist	50000
auto_stmt_stats	ON

- Set the tablespace pageSize parameter to 32768 bytes.

  In a single-partition database, specify a tablespace that meets the pageSize requirements. If you do not specify a tablespace, the default tablespace must meet the pageSize requirements.

  In a multi-partition database, specify a tablespace that meets the pageSize requirements. Define the tablespace in the catalog partition of the database.
- Set the NPAGES parameter to at least 5000. The NPAGES parameter determines the number of pages in the tablespace.
- Verify that the database user has CREATETAB, CONNECT, and BINDADD privileges.
- Informatica does not support IBM DB2 table aliases for repository tables. Verify that table aliases have not been created for any tables in the database.
- In the DataDirect Connect for JDBC utility, update the DynamicSections parameter to 3000.

  The default value for DynamicSections is too low for the Informatica repositories. Informatica requires a larger DB2 package than the default. When you set up the DB2 database for the domain configuration repository or a Model repository, you must set the DynamicSections parameter to at least 3000. If the DynamicSections parameter is set to a lower number, you can encounter problems when you install or run Informatica services.

Microsoft SQL Server Database Requirements

Use the following guidelines when you set up the repository on Microsoft SQL Server:

- Set the read committed isolation level to READ_COMMITTED_SNAPSHOT to minimize locking contention.
  To set the isolation level for the database, run the following command:

  ```
  ALTER DATABASE DatabaseName SET READ_COMMITTED_SNAPSHOT ON
  ```

  To verify that the isolation level for the database is correct, run the following command:

  ```
  SELECT is_read_committed_snapshot_on FROM sys.databases WHERE name = DatabaseName
  ```
- Set the ALLOW_SNAPSHOT_ISOLATION parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET ALLOW_SNAPSHOT_ISOLATION ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_allow_snapshot_isolation_on FROM sys.databases WHERE name = DatabaseName
  ```
- Set the AUTO_UPDATE_STATISTICS_ASYNC parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS_ASYNC ON
  ```

To verify that the parameter is set correctly, run the following command:

```
SELECT is_auto_update_statistics_async_on FROM sys.databases WHERE name = DatabaseName
```

- Set the AUTO_UPDATE_STATISTICS parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_on FROM sys.databases WHERE name = DatabaseName
  ```

- The database user account must have the CONNECT, CREATE TABLE, and CREATE VIEW privileges.

Oracle Database Requirements

Use the following guidelines when you set up the repository on Oracle:

- Set the open_cursors parameter to 4000 or higher.
- Verify that the database user has the CONNECT, RESOURCE, and CREATE VIEW privileges.
- Informatica does not support Oracle public synonyms for repository tables. Verify that public synonyms have not been created for any tables in the database.

Profiling Warehouse Database Requirements

The profiling warehouse database stores profiling and scorecard results. You specify the profiling warehouse connection when you create the Data Integration Service.

The profiling warehouse supports the following database types:

- IBM DB2 UDB
- Microsoft SQL Server
- Oracle

Allow 10 GB of disk space for the database.

Note: Ensure that you install the database client on the machine on which you want to run the Data Integration Service. You can specify a JDBC connection or Hive connection as a profiling warehouse connection for IBM DB2 UDB, Microsoft SQL Server, and Oracle database types. You can create column profiles, rule profiles, data domain discovery profiles, and scorecards with a JDBC connection as a profiling warehouse connection.

For more information about configuring the database, see the documentation for your database system.

IBM DB2 Database Requirements

Use the following guidelines when you set up the repository on IBM DB2:

- The database user account must have the CREATETAB, CONNECT, CREATE VIEW, and CREATE FUNCTION privileges.
- Informatica does not support IBM DB2 table aliases for repository tables. Verify that table aliases have not been created for any tables in the database.
- Set the tablespace pageSize parameter to 32768 bytes.
- Set the NPAGES parameter to at least 5000. The NPAGES parameter determines the number of pages in the tablespace.

Note: Informatica does not support the partitioned database environment for IBM DB2 databases when you use a JDBC connection as the profiling warehouse connection.

Microsoft SQL Server Database Requirements

Use the following guidelines when you set up the repository on Microsoft SQL Server:

- Set the read committed isolation level to READ_COMMITTED_SNAPSHOT to minimize locking contention. To set the isolation level for the database, run the following command:

  ```
  ALTER DATABASE DatabaseName SET READ_COMMITTED_SNAPSHOT ON
  ```

  To verify that the isolation level for the database is correct, run the following command:

  ```
  SELECT is_read_committed_snapshot_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the ALLOW_SNAPSHOT_ISOLATION parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET ALLOW_SNAPSHOT_ISOLATION ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_allow_snapshot_isolation_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the AUTO_UPDATE_STATISTICS_ASYNC parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS_ASYNC ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_async_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the AUTO_UPDATE_STATISTICS parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_on FROM sys.databases WHERE name = DatabaseName
  ```

- The database user account must have the CONNECT, CREATE TABLE, CREATE VIEW, and CREATE FUNCTION privileges.

Oracle Database Requirements

Use the following guidelines when you set up the repository on Oracle:

- Verify that the database user has the following privileges:

  ALTER TABLE

  CREATE ANY INDEX

  CREATE PROCEDURE

  CREATE SESSION

  CREATE TABLE

  CREATE VIEW

  DROP TABLE

  UPDATE TABLE

- Informatica does not support Oracle public synonyms for repository tables. Verify that public synonyms have not been created for any tables in the database.
- Set the tablespace parameter. Use the following formula to determine the value: 2 MB x (number of tables in each scan x number of concurrent scans)
  For example, you have 1,000 tables in each scan and you plan to run 10 scans concurrently. Calculate the tablespace parameter value as follows: 2 MB x (100 x 10) = 20 GB

  Note: Tablespace must be distributed across multiple disks.

- Set the following parameters to the Informatica recommended values:

Parameter	Recommended Value
open_cursors	3000
Sessions	1000
Processes	1000

Operating System Parameters

Set the following parameters to the Informatica recommended values:

Parameter	Recommended Value
descriptors	65535
maxproc	65535
stacksize	50000 KB

Reference Data Warehouse Database Requirements

The reference data warehouse stores the data values for reference table objects that you define in a Model repository. You configure a Content Management Service to identify the reference data warehouse and the Model repository.

You associate a reference data warehouse with a single Model repository. You can select a common reference data warehouse on multiple Content Management Services if the Content Management Services identify a common Model repository. The reference data warehouse must support mixed-case column names.

The reference data warehouse supports the following database types:

- IBM DB2 UDB
- Microsoft SQL Server
- Oracle

Allow 200 MB of disk space for the database.

Note: Ensure that you install the database client on the machine on which you want to run the Content Management Service.

IBM DB2 Database Requirements

Use the following guidelines when you set up the repository on IBM DB2:

- Verify that the database user account has CREATETAB and CONNECT privileges.
- Verify that the database user has SELECT privileges on the SYSCAT.DBAUTH and SYSCAT.DBTABAUTH tables.
- Informatica does not support IBM DB2 table aliases for repository tables. Verify that table aliases have not been created for any tables in the database.
- Set the tablespace pageSize parameter to 32768 bytes.
- Set the NPAGES parameter to at least 5000. The NPAGES parameter determines the number of pages in the tablespace.

Microsoft SQL Server Database Requirements

Use the following guidelines when you set up the repository on Microsoft SQL Server:

- Set the read committed isolation level to READ_COMMITTED_SNAPSHOT to minimize locking contention. To set the isolation level for the database, run the following command:

  ```
  ALTER DATABASE DatabaseName SET READ_COMMITTED_SNAPSHOT ON
  ```

  To verify that the isolation level for the database is correct, run the following command:

  ```
  SELECT is_read_committed_snapshot_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the ALLOW_SNAPSHOT_ISOLATION parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET ALLOW_SNAPSHOT_ISOLATION ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_allow_snapshot_isolation_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the AUTO_UPDATE_STATISTICS_ASYNC parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS_ASYNC ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_async_on FROM sys.databases WHERE name = DatabaseName
  ```

- Set the AUTO_UPDATE_STATISTICS parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_on FROM sys.databases WHERE name = DatabaseName
  ```

- The database user account must have the CONNECT and CREATE TABLE privileges.

Oracle Database Requirements

Use the following guidelines when you set up the repository on Oracle:

- Verify that the database user has the following privileges:

  ALTER SEQUENCE

  ALTER TABLE

  CREATE SEQUENCE

  CREATE SESSION

  CREATE TABLE

  CREATE VIEW

  DROP SEQUENCE

  DROP TABLE

- Informatica does not support Oracle public synonyms for repository tables. Verify that public synonyms have not been created for any tables in the database.

Secure@Source Repository Database Requirements

Secure@Source stores data and metadata, such as data stores and scans, in the Secure@Source repository. Before you create the Secure@Source Service, set up a database and database user account for the Secure@Source repository.

The Secure@Source repository supports the following database types:

- IBM DB@ UDB
- Microsoft SQL Server
- Oracle

Allow 3 GB of disk space for DB2. Allow 200 MB of disk space for all other database types.

For more information about configuring the database, see the documentation for your database system.

IBM DB2 Database Requirements

Use the following guidelines when you set up the repository on IBM DB2:

- If the repository is in an IBM DB2 9.7 database, verify that IBM DB2 Version 9.7 Fix Pack 7 or a later fix pack is installed.
- On the IBM DB2 instance where you create the database, set the following parameters to ON:
  - DB2_SKIPINSERTED
  - DB2_EVALUNCOMMITTED
  - DB2_SKIPDELETED
  - AUTO_RUNSTATS
- On the database, set the configuration parameters.

  The following table lists the configuration parameters that you must set:

Parameter	Value
applheapsz	8192
appl_ctl_heap_sz	8192
logfilsiz	8000
maxlocks	98
locklist	50000
auto_stmt_stats	ON

- Set the tablespace pageSize parameter to 32768 bytes.

  In a single-partition database, specify a tablespace that meets the pageSize requirements. If you do not specify a tablespace, the default tablespace must meet the pageSize requirements.

  In a multi-partition database, specify a tablespace that meets the pageSize requirements. Define the tablespace in the catalog partition of the database.
- Set the NPAGES parameter to at least 5000. The NPAGES parameter determines the number of pages in the tablespace.
- Verify that the database user has CREATETAB, CONNECT, and BINDADD privileges.

- Informatica does not support IBM DB2 table aliases for repository tables. Verify that table aliases have not been created for any tables in the database.
- In the DataDirect Connect for JDBC utility, update the DynamicSections parameter to 3000.

  The default value for DynamicSections is too low for the Informatica repositories. Informatica requires a larger DB2 package than the default. When you set up the DB2 database for the domain configuration repository or a Model repository, you must set the DynamicSections parameter to at least 3000. If the DynamicSections parameter is set to a lower number, you can encounter problems when you install or run Informatica services.

Microsoft SQL Server Database Requirements

Use the following guidelines when you set up the repository on Microsoft SQL Server:

- Set the read committed isolation level to READ_COMMITTED_SNAPSHOT to minimize locking contention.
  To set the isolation level for the database, run the following command:

  ```
  ALTER DATABASE DatabaseName SET READ_COMMITTED_SNAPSHOT ON
  ```

  To verify that the isolation level for the database is correct, run the following command:

  ```
  SELECT is_read_committed_snapshot_on FROM sys.databases WHERE name = DatabaseName
  ```
- Set the ALLOW_SNAPSHOT_ISOLATION parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET ALLOW_SNAPSHOT_ISOLATION ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_allow_snapshot_isolation_on FROM sys.databases WHERE name = DatabaseName
  ```
- Set the AUTO_UPDATE_STATISTICS_ASYNC parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS_ASYNC ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_async_on FROM sys.databases WHERE name = DatabaseName
  ```
- Set the AUTO_UPDATE_STATISTICS parameter to ON.
  To set the parameter to ON, run the following command:

  ```
  ALTER DATABASE DatabaseName SET AUTO_UPDATE_STATISTICS ON
  ```

  To verify that the parameter is set correctly, run the following command:

  ```
  SELECT is_auto_update_statistics_on FROM sys.databases WHERE name = DatabaseName
  ```
- The database user account must have the CONNECT, CREATE TABLE, and CREATE VIEW privileges.

Oracle Database Requirements

Use the following guidelines when you set up the repository on Oracle:

- Set the SESSIONS parameter to 1500.
- Set the PROCESSES parameter to 2272.
- Set the OPEN_CURSORS parameter to 2000 or higher.
- Verify that the database user has the CONNECT, RESOURCE, and CREATE VIEW privileges.
- Informatica does not support Oracle public synonyms for repository tables. Verify that public synonyms have not been created for any tables in the database.

Database User Privileges for Jobs

Secure@Source jobs require specific privileges for the user account that connects to the source. For example, the Informatica PowerCenter scan job and the database scan job require specific privileges to connect to and run scans on databases. Before you scan a data store or run an import, verify that the user account configured in the data store or connection has the required privileges. If the user account does not have the required privileges, the job fails.

User Privileges for the Cloudera Navigator Scan Job

The Cloudera Navigator scan job requires a Cloudera Manager user account that includes the Full Administrator or Navigator Administrator role. The roles include authorization to connect to and retrieve metadata from Cloudera Navigator. If the user account does not have one of these roles, the scan job fails.

Database User Privileges for the Database Scan Job

The database scan job requires user account privileges to connect to and run a scan on a database. When you create a data store, you specify the user account. When an Informatica PowerCenter scan job imports data stores, the user account is populated from the connection attributes in the PowerCenter repository.

Before you run a database scan on a data store, verify that the user account in the data store has the required privileges. The required privileges depend on the database type.

IBM DB2 Database User Privileges

Verify the permissions for the database user account that you use to connect to an IBM DB2 database.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the IBM DB2 database user account:

Schema Objects	System Tables
Tables Views Indexes Packages Procedures Functions Sequences Triggers Synonyms	SYSCAT.SCHEMATA SYSCAT.TRIGGERS SYSCAT.INDEXES SYSCAT.INDEXCOLUSE SYSCAT.REFERENCES SYSCAT.TABCONST SYSCAT.KEYCOLUSE SYSCAT.COLCHECKS SYSCAT.DATATYPES SYSCAT.COLUMNS SYSCAT.TABLES SYSCAT.FUNCTIONS SYSCAT.PROCEDURES SYSCAT.PACKAGES SYSCAT.TRIGGERS SYSCAT.SEQUENCES SYSCAT.INDEXES SYSCAT.CHECKS SYSCAT.VIEWS SYSCAT.PACKAGES SYSCAT.FUNCDEP SYSCAT.TRIGDEP SYSCAT.VIEWDEP

In addition, the database user account must have CREATEIN, DROPIN, and ALTERIN privileges on the NULLID schema. Optionally, you can grant the BINDADD privilege to the user account.

IBM DB2 for z/OS Database User Privileges

Verify the permissions for the database user account that you use to connect to an IBM DB2 for z/OS database.

Configure SELECT privileges on the following tables in the DB2 for z/OS user account that you use to connect to the metadata source:

System Tables	Metadata Extracted
SYSIBM.SYSSYNONYMS SYSIBM.SYSTABLES SYSIBM.SYSROUTINES SYSIBM.SYSSEQUENCES SYSIBM.SYSDATATYPES	Schemas
SYSIBM.SYSTABLES SYSIBM.SYSTABLESPACE SYSIBM.SYSVIEWS	Tables Views Alias
SYSIBM.SYSTABLES SYSIBM.SYSCOLUMNS SYSIBM.SYSTABLESPACE	Columns
SYSIBM.SYSKEYS SYSIBM.SYSINDEXES	Primary keys
SYSIBM.SYSRELS SYSIBM.SYSFOREIGNKEYS SYSIBM.SYSINDEXES	Foreign keys
SYSIBM.SYSCHECKS SYSIBM.SYSCHECKDEP	Check constraints
SYSIBM.SYSINDEXES SYSIBM.SYSKEYS	Indexes
SYSIBM.SYSSYNONYMS SYSIBM.SYSTABLES	Synonyms
SYSIBM.SYSTRIGGERS	Triggers
SYSIBM.SYSROUTINES	Stored procedures
SYSIBM.SYSROUTINES	Functions
SYSIBM.SYSSEQUENCES	Sequences
SYSIBM.SYSSEQUENCESDEP SYSIBM.SYSSEQUENCES	Identity
SYSIBM.SYSDATATYPES	User-defined types

Note: If the PowerExchange Listener does not use DB2 security, configure SELECT privileges for the PowerExchange Listener user ID instead of the DB2 for z/OS user account.

Microsoft SQL Server Database User Privileges

Verify the permissions for the database user account that you use to connect to a Microsoft SQL Server database.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the Microsoft SQL Server database user account:

Schema Objects	System Tables
Tables Views Indexes Packages Procedures Functions Sequences Triggers Synonyms	sysobjects sysusers master.dbo.sysservers syscolumns sysindexkeys systypes syscomments sysreferences sysdepends sys.synonyms (Microsoft SQL Server 2005) sysindexes sys.sql_expression_dependencies

In addition, the user must have VIEW DEFINITION permission on the database.

Netezza Database User Requirements

Verify the permissions for the database user account that you use to connect to a Netezza database. You specify the user account in the data store connection properties.

The user account must have the LIST privilege to extract metadata from a Netezza database.

Configure SELECT permissions for the Netezza user account on the following system views:

- _V_TABLE_DIST_MAP
- _V_TABLE
- _V_DATABASE
- _v_relation_keydata
- _v_relation_column
- _v_view
- _v_synonym
- _v_datatype
- _v_sequence

Oracle Database User Privileges

Verify the permissions for the database user account that you use to connect to an Oracle database. You specify the user account in the data store connection properties.

Configure CONNECT and SELECT_CATALOG_ROLE permissions for the user account.

If you cannot configure SELECT_CATALOG_ROLE permission for the user account, configure the following permissions:

- SELECT permissions on the v$version and v$instance views.

- CONNECT and SELECT permissions on the required schema objects and corresponding system tables.

Configure SELECT permissions on the following schema objects:

- Tables
- Views
- Indexes
- Packages
- Procedures
- Functions
- Sequences
- Triggers
- Synonyms

Configure SELECT permissions on the following system tables:

- ALL_OBJECTS
- ALL_SOURCE
- ALL_USERS
- DBA_COL_COMMENTS
- DBA_CONS_COLUMNS
- DBA_CONSTRAINTS
- DBA_DEPENDENCIES
- DBA_IND_COLUMNS
- DBA_INDEXES
- DBA_MVIEWS
- DBA_OBJECTS
- DBA_RECYCLEBIN
- DBA_SEQUENCES
- DBA_SOURCE
- DBA_SYNONYMS
- DBA_TAB_COLUMNS
- DBA_TAB_COMMENTS
- DBA_TAB_PARTITIONS
- DBA_TABLES
- DBA_TRIGGERS
- DBA_VIEWS
- SYS.V_$DATABASE
- SYS.DBA_USERS
- SYS.DBA_DIRECTORIES
- SYS.DBA_EXTERNAL_LOCATIONS
- SYS.DBA_CLU_COLUMNS

If schemas have a table with a cluster defined on some columns, you must configure SELECT permission on the DBA_CLUSTER system table.

Sybase Database User Privileges

Verify the permissions for the database user account that you use to connect to a Sybase database.

Configure SELECT permissions on the following schema objects:

- Tables
- Views
- Columns
- Procedures

Configure SELECT permissions on the following system tables:

- sysobjects
- syscomments
- sysdepends
- syscolumns
- sysindexes
- sysconstraints
- sysreferences
- sysusers
- systypes
- master.dbo.sysservers
- master.dbo.sysconfigures
- @@<servername>

Teradata Database User Privileges

Verify the permissions for the database user account that you use to connect to a Teradata database. You specify the user account in the data store connection properties.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the Teradata database user account:

Schema Objects	System Tables
Tables Views Indexes Packages Procedures Functions Sequences Triggers Synonyms	DBC.ALL_RI_CHILDREN DBC.COLUMNS DBC.DATABASES DBC.DBASE DBC.DBCINFO DBC.HOSTSINFO DBC.INDICES DBC.SHOWTBLCHECKS DBC.TABLES DBC.TABLETEXT DBC.TRIGGERS DBC.TVFIELDS DBC.TVM DBC.UDTINFO TDQCD.STATSRECS

User Privileges for the Big Data Scan Job

The Big Data scan job requires the following privileges:

- The Big Data scan job requires a user account that has read, write, and execute privileges on the directory configured in the Hive Warehouse Directory On HDFS data store property. If the user account does not have the required privileges, the scan job fails.

  You specify the user in the Hive data store connection properties.
- By default, the Big Data scan job uses the Blaze engine to run mappings in the Hadoop cluster. The Blaze user must have permissions to run mappings in the Hadoop cluster. You specify the Blaze user when you run the Hadoop Configuration Manager to create a Big Data connection in the Informatica domain.

Database User Privileges for the Informatica PowerCenter Scan Job

The Informatica PowerCenter scan job requires database privileges to connect to and run a scan on a PowerCenter repository. The scan job also requires privileges for the PowerCenter user that logs in to the PowerCenter client tools. You specify the users when you create a data store that connects to Informatica PowerCenter.

Before you run an Informatica PowerCenter scan on a data store, verify that the user account and the PowerCenter user have the required privileges. If the users do not have the required privileges, the scan job fails.

PowerCenter Repository Database User Account

The PowerCenter repository database user account must have the CREATE_VIEW privilege. If you use a read-only user, the user account must have the CREATE_SYNONYM privilege.

PowerCenter User Account

The PowerCenter user account logs in to the PowerCenter client tools and accesses the PowerCenter repository.

The Informatica PowerCenter scan job uses the PowerCenter user account to create a folder in the PowerCenter repository. The folder includes a workflow that reads connection properties from database configuration files and PowerCenter parameter files. The workflow is required for the scan job to import the complete connection properties for all of the connections in the PowerCenter repository.

The PowerCenter user account must have PowerCenter Repository Service privileges for the following privilege groups:

- Folders
- Runtime objects
- Sources and targets
- Design objects

For more information about the privilege groups, see the *Informatica Security Guide*.

User Privileges for Salesforce Import Jobs

The Salesforce import job requires org and user account privileges to connect to and import user metadata from Salesforce. You specify the user account in the Salesforce data store connection properties. Before you import user metadata from Salesforce, verify that the user account in the data store has the required privileges.

The Salesforce import job requires the following privileges:

- The Salesforce org must have the Bulk API feature enabled.
- The user account must have the View All Data and the View Event Log Files privileges.

Author